

REALTIQ: Feature Extraction & Architectural Categorisation

Based on the strategic document provided, the features and ideations have been categorised into functional product modules. This structure maps directly to how we will architect the database schemas and micro-services. I am still not finished with what to monetise.

1. Financial & Transaction Engine (The "Checkout" Module)

This module handles all money movement, contract logic, and platform monetisation.

- **Escrow Account Engine:** Condition-based payment locking and release; custom rule definitions for individual buyers/sellers.
- **Instalment Payments:** Smart-contract or milestone-based payment schedules.
- **Outright Payments:** Direct fiat processing.
- **Ideation (Pending):** Monetisation strategies (fee structures, SaaS models, premium visibility, etc).

2. Identity, Trust & Security Module

This module handles user verification and reputation management.

- **The "Trust" Badge System:** Tiered verification badges.
- **Vendor/Seller/Owner Ratings:** Peer and transaction-based scoring.
- **Trustworthiness Rating:** Algorithmic trust scoring.
- **Cryptographic Title Hashing (NEW MOAT):** To combat "dual-selling" (selling the same land to two people), once a property's title is verified by our legal team, we generate a cryptographic hash of the document and timestamp it on a public ledger. This creates an immutable, undeniable record of who listed what, and when.
- **Independent Diaspora Proxy Network (NEW MOAT):** Diaspora buyers are the highest-value users but face the highest risk. We will build an "Uber for Inspectors" feature. A diaspora buyer can pay a micro-fee to dispatch a REALTIQ-verified, independent professional to the site to do a live video walkthrough and sign a condition report before the buyer releases escrow.

3. Spatial Visualisation & Immersive Media

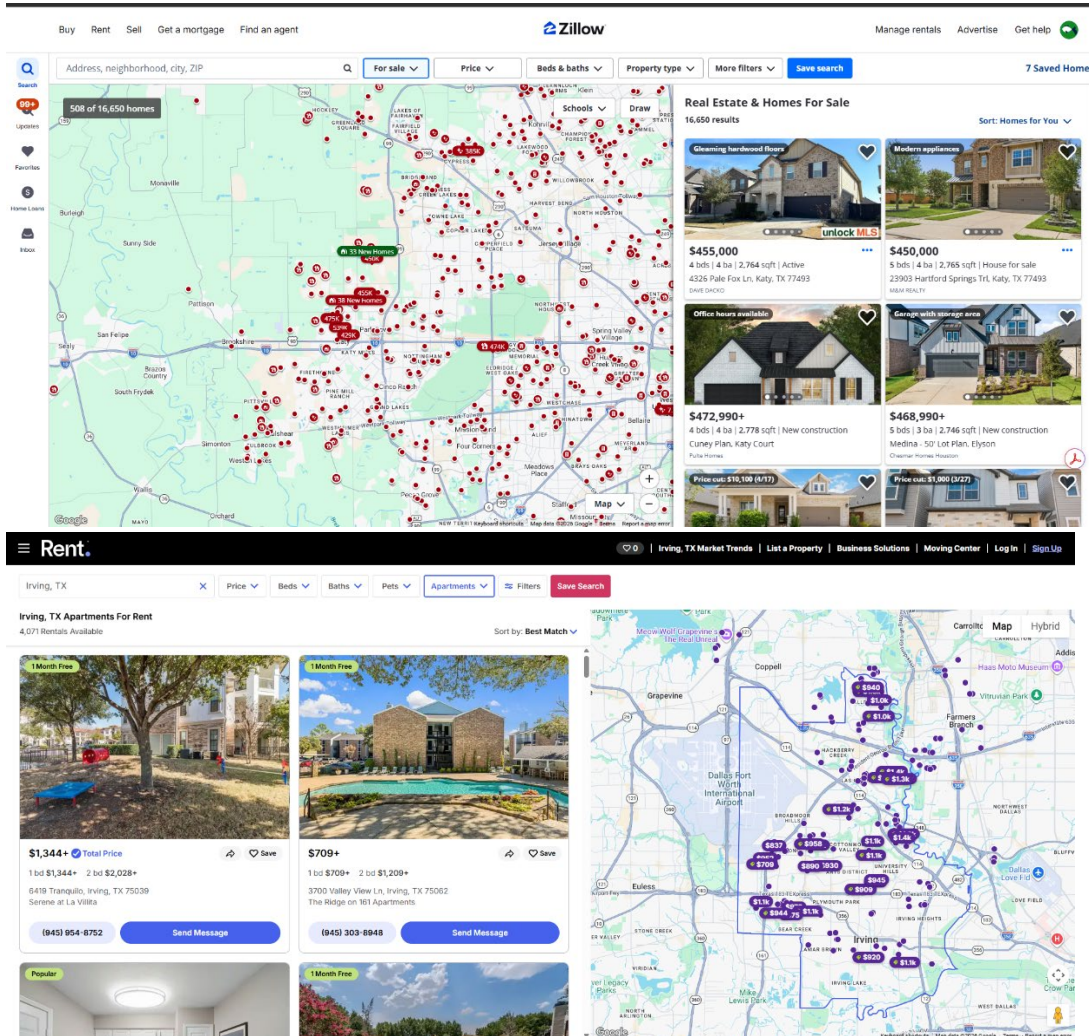
This module governs the high-fidelity property viewing experience. It requires strict data-optimisation protocols.

- ***If possible, at the initial stage* Native Immersive Media Viewer:** A proprietary, flawless WebGL native viewer (eliminating reliance on external Matterport/Realsee iframes). *If too expensive or cumbersome or not possible initially, we use Matterport or Realsee*.
- **3D Models:** 3D virtual tours and 3D Dollhouse views.

- **Tour Variations:** Open house tours, Paid virtual tours, Staging views.
- **2D Visuals:** Photos, Basic and Interactive Floorplans.
- **Aerial Data:** Drone shots.

4. Map-First Discovery Engine (Geo-Spatial Module)

This shifts the UI from list-views to a Zillow-style spatial discovery experience, leveraging Mapbox/Google Maps APIs (PostGIS required on backend). I'd like an interactive map search feature that works exactly like the Zillow or Rent.com maps. Basically, the map should feel like a living search tool.



- **Interactive Map-Based Discovery:** The core search interface.
 - **The Map Controls the List:** The map and the list of properties next to it need to be linked. Whenever a user zooms in, zooms out, or drags the map to a new area, the list of properties on the side should automatically update to only show the homes that are currently visible on the screen.
 - **Filters Work on Both at Once:** If a user uses the search filters (like changing the price, or selecting "2 bedrooms"), both the dots on the map and the list of homes

should update at the exact same time.

- **Connected Highlighting:** *If a user hovers over a property in the side list, the matching dot on the map should highlight or bounce so they know where it is. Similarly, clicking a dot on the map should pull up that specific property in the list.*
- **Grouping for Cleanliness:** *If a user zooms way out so that it covers a huge area, the map shouldn't get cluttered with hundreds of individual dots. Instead, they should group together into a single circle that just shows a number (like "50 listings in this area"). When they zoom back in, it spreads back out into individual dots.*
- **Map Types & Overlays:** Basemaps, dynamic/static overlays, branded maps, high-res map capture, masterplans.
- **Premium Map Features:** Paid Streetview, Site plans, Masterplans (Paid).
- **Advanced Spatial Data:** Lidar scanning integration.
- **Advanced Spatial Heat Map Engine (Proprietary Data Layers):**
 - *Market Activity & Demand Overlays:* Search density, Listing density, Transaction velocity.
 - *Development & Construction Overlays:* Active development zones, Construction supply drop zones.
 - *Financial & Yield Overlays:* Price per SQM hotspots, Highest rental yield zones, Capital appreciation projections.
 - *Risk & Environmental Overlays:* Flood risk zones/Topography, Reported security incident density.
 - *Infrastructure Overlays:* Proximity to stable power grids, Broadband internet coverage, Paved road networks.
 - *Dynamic Heat Map Filters:* Building Type, User Persona, Timeframe, Property Tier.
- **Progressive Web App (PWA) Offline Mode (NEW MOAT):** Internet connectivity is highly variable on active construction sites or remote land plots. The mobile web app will cache map data and let field agents/buyers save properties for offline viewing. Actions taken offline (like signing a contract or sending a message) will sync to the server the moment the device hits a 3G/4G/WiFi network.

5. Property Data & Document Vault

This handles static listing data and secure document distribution.

- **Standard Property Features:** Type, Architectural Style, Details, Living space (area), Property space (plot area), Construction details (e.g., year built), Neighbourhood.
- **Document Management System:** "Pay-to-view" secure legal document vault (monetised micro-transaction for title documents, surveys, etc.).
- **Automated E-Contracts & Digital Signatures (NEW):** Do not let the transaction leave the platform for paperwork. We will auto-generate standard Nigerian Sales Agreements and Tenancy Agreements, integrating local e-signature APIs to make them legally binding instantly.

6. User Workflow & Communications

This handles the pipeline from interest to transaction initiation.

- **Purchase Flows:** Distinct, optimised UI/UX workflows for Buyers vs. Sellers (Developers/Property Owners).
- **Internal Messaging:** Built-in chat system for buyer/seller negotiations.
- **Tour Management:** Booking, scheduling, and requesting physical viewing tours.
- **Document Requests:** Workflows to request restricted title documents.

7. Data Intelligence & Automated Valuation (AVM)

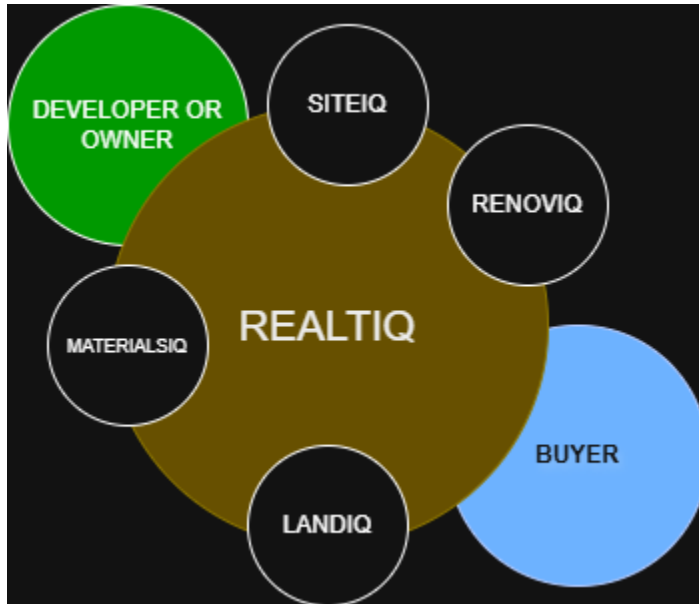
This is the core analytics engine, ingesting data from multiple sources to provide market intelligence.

- **ROI Calculators:** Specific interfaces for Buyers and Sellers.
- **Proprietary Valuation Model (Estimated Market Value):**
 - *Core Inputs:* Land cost, material cost, construction cost, comparables, living space, plot size.
 - *Ecosystem Inputs:* Live pricing (MATERIALIQ), construction progress/BIM (SITEIQ), lifecycle costs (RENOVIO), contractor pricing bids (KONSTRUCTIQ).
 - *Macro/Micro Indicators:* Infrastructure, accessibility, security, supply/demand, rental yield, zoning policies, inflation, interest/mortgage rates, demographic trends, environmental risks (floods, etc.), property age, design quality, energy efficiency, historical price appreciation.
- **Hyper-Local Yield vs. Inflation Tracker (NEW):** An analytics dashboard specifically showing if a property's historical appreciation outpaces the local inflation rate (a critical metric for institutional investors in Nigeria).
- **Market Data:** Price history, Macro/Micro info, Industry news, Regulatory/Government news.
- **Listing Analytics:** View counts, Save counts.

8. Ecosystem Architecture & Integrations

These are the planned API tie-ins to our broader PropTech ecosystem.

- **User Types:** Direct to Consumer (Companies and Individuals).
 - **SELLER – DEVELOPER or OWNER** (Corporate and Individuals)
 - **BUYER** (Corporate and Individuals)
 - **REALTIQ**



- **MATERIALIQ:** Integration for construction material supply/pricing.
- **SITEIQ:** Integration for remote quality control, BIM, and reality capture.
- **RENOVIO:** Integration for facility management and lifecycle maintenance data.
- **KONSTRUCTIQ:** Integration for construction contracting and engineering bids.
- **LANDIQ:** (Noted from the visual diagram) Future integration for land data.

9. User Engagement & Retention (The "CRM" Module)

Standard quality-of-life features for account holders.

- **Personalisation:** Favourites, Saved searches, and Recently viewed.
- **Notifications:** Listing updates, recently added properties.
- **Content:** Blog and educational site content.